An fMRI-Guided Deep Learning Framework for Rapid Semi-Automated Deep Brain Stimulation Optimization in Parkinson's Disease

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***Abstract*—While it takes a lot of time to optimize parameters for deep brain stimulation (DBS) in Parkinson's disease (PD), it is beneficial in therapeutic environments and typically necessitates many appointments over the course of around a year. The use of directional electrodes and slower clinical feedback makes this challenge more difficult. We recommend a novel technique that combines fMRI and deep learning to quickly optimize DBS parameters during a single clinical session. To allow it to recognize significant neuronal features, an unsupervised autoencoder (AE) was trained on 122 BOLD-fMRI datasets from 39 PD patients whose therapy was carefully adjusted. MLP models next utilized these characteristics to forecast and categorize DBS parameters. The AE was able to clearly distinguish the most from the least successful stimulation states. With an area under the curve (AUC) of 0.98, the AE-MLP classifier attained an accuracy rate of 0.96 ± 0.04, a precision of 0.95 ± 0.07, a recall of 0.92 ± 0.07, and an F1-score of 0.93 ± 0.06. The voltage prediction accuracy was 0.79 0.04, the frequency prediction accuracy was 0.85 0.04, and the x-y-z contact locations had an accuracy of 0.70 to 0.83, where x-y-z are the electrode contacts' MNI coordinates. These results indicate that semi-automated DBS optimization utilizing fMRI is feasible and might significantly reduce the time required for optimization, lower patient expenses, and decrease the burden on medical professionals.**

Keywords—Deep brain stimulation optimization; Parkinson's disease; Deep learning; fMRI; Autoencoder; Unsupervised feature extraction; Multilayer perceptron; Neurostimulation

I. Introduction

Deep brain stimulation (DBS) is a type of therapy that uses implanted electrodes to transmit continuous electrical signals to change abnormal brain behavior. Parkinson's disease, essential tremor, and dystonia are just a few of the movement disorders that DBS is often used to treat. It has also proven helpful in treating epilepsy, pain disorders, cognitive deficits, and psychiatric illnesses. The efficacy of DBS as a treatment for Parkinson's disease, which affects areas such the subthalamic nucleus and globus pallidus internus, depends on the proper placement of electrodes and the tailoring of stimulation parameters. Customized programming enhances health advantages and reduces negative side effects, whereas improper settings can result in lower efficiency, more adverse effects, and quicker battery drain [1].

Since the current approach to DBS programming necessitates several hands-on adjustments over the course of several clinic appointments, it can take a patient up to a year on average to achieve the ideal settings. Due to this lengthy process, the patient suffers significant financial and bodily hardship. The introduction of directional electrodes has complicated the parameters that need to be modified, making the traditional optimization approach less feasible and impeding the widespread use of advanced DBS solutions. The situation is made more complicated by conditions like dystonia, addiction, and depression, where the effects of It could take weeks for the effects of environmental shifts to become apparent. For this reason, the total number of patients worldwide who have had DBS therapy—more than 230,000 people—is still relatively low in comparison to the number of people who may truly benefit from it.

Using biomarkers to optimize DBS is one method for significantly shortening the time it takes for patients to reach beneficial settings. Previous studies have shown that the best DBS parameters for Parkinson's patients produce distinct fMRI response patterns that are contingent upon the amount of blood oxygenation. Lower and higher-than-effective stimulation causes particular variations in the strength and area of the response, but the left subthalamic nucleus is best stimulated, producing unique patterns of activation and deactivation in different parts of the brain. Using these distinctive DBS-fMRI patterns, it's possible to identify traits that point to optimal stimulation levels.

Taking this idea as a foundation, we propose a deep learning-based technique for quickly and semi-automatically modifying DBS parameters. A self-supervised autoencoder (AE) will be used to extract underlying features from DBS-fMRI response images without the need for predefined anatomical divisions. The multilayer perceptron (MLP) models for identifying and predicting DBS parameters will subsequently be developed using these features. Additionally, we will examine the AE's capacity to differentiate between ideal and non-ideal stimulation settings, taking into account variations in the stimulation site and the nature of the condition, so that DBS optimization may be approached more effectively and holistically [2].

II. Methods And Process

All MRI tests were carried out using DBS systems approved for use in an MRI context, following the maker's safety recommendations. The patient was closely watched throughout the imaging process maintaining the particular absorption rate limitations for safety. The objects connected with DBS were restricted to the area surrounding the electrodes and did not influence the majority of the cortical and subcortical areas needed for feature learning.

A. Dataset Explanation

We used 39 Parkinson's disease (PD) patients (n = 35 STN- DBS, n = 4 GPi- DBS; mean age = 62.4 ± 7.1 years; 20 men, 19 women; clinical trial ID: NCT03153670) who had left-sided DBS implantation at Toronto Western Hospital; their previously acquired BOLD fMRI data collected on a 3T GE HDx MRI scanner (GE Healthcare, Wisconsin, USA) was used. Previously, the standard-of-care clinical optimization technique helped to define ideal DBS settings for every patient. Under several DBS settings, fMRI data were next gathered for a total of 122 DBS-fMRI reaction maps [3].

The University Health Network Research Ethics Board (REB #14-8255) approved all study plans. All patients gave written informed consent before MRI scanning; a member of the clinical team was present throughout each imaging session to monitor them. Patients stopped PD medicines at least 24 hours before to scanning to reduce confounds related to pharmaceuticals. Previously reported are more information on patient demographics and methods of data collecting.

To distinguish patterns of brain activity related with ideal and non-optimal DBS settings, the experimental design of DBS-fMRI was improved. Using a 30-second DBS ON/OFF block paradigm (Fig. 1), each DBS-fMRI session took 6.5 minutes. Each response map was designated as ideal or non-optimal by a movement disorder specialist depending on the clinically established optimum settings, which linked to a unique DBS parameter configuration. For rigid-body registration, high-resolution axial 3D T1-weighted anatomical pictures were gathered and used. Additional information on the fMRI and anatomical imaging settings is found in [4].

Although the cohort size is small (39 patients, 122 DBS-fMRI datasets), this captures the realistic limits of getting DBS-fMRI data in implanted patients under various stimulation settings. Still a very specialized and expensive process, DBS-fMRI has one of the largest clinically optimized DBS-fMRI cohorts now available in this dataset. Still, we realize that the somewhat small sample size and the predominance of left-sided STN stimulation might restrict broad generalizability to more varied PD populations, GPi targets, and other stimulation approaches [5].

B. Parameter sampling based on DBS

Using a methodical method for sampling DBS variables, the DBS-fMRI data was collected. Using clinically optimized DBS parameters that were previously determined using the standard of care clinical methodology, fMRI data on the optimum stimulation were collected for each patient. Among the characteristics of these were the voltage, frequency, pulse width, and position of the electrode contact. Different electrode contact sites that the patient could tolerate, voltages outside the range of therapeutic values, and stimulation frequencies between 80 and 160 Hz were utilized in additional DBS-fMRI data collection to generate non-optimal stimulation configurations. The DBS pulse duration for all 39 individuals was set at 60 microseconds. The number of DBS parameter setups tested for each patient varied between one and five, depending on individual tolerance, as well as the number of DBS-fMRI data sets collected for each patient. Because the acceptable subtherapeutic and supratherapeutic parameter pairings varied from patient to patient, the resulting non-ideal DBS parameter combinations were not consistent throughout the cohort.

C. Preparing for an MRI

Step	Description
Slice-timing correction	Corrected for interleaved slice acquisition
Motion correction	ART toolbox; volumes > 2 mm displacement removed
Coregistration	Functional - T1 anatomical
Normalization	Nonlinear normalization to MNI152 space
Spatial Smoothing	6-mm FWHM gaussian kernel
Temporal filtering	Implicit high-pass filtering via GLM (128 s cutoff)
Global signal regression	Not applied, consistent with prior DBS-fMRI studies

Table. 1 Overview of fMRI Preparation Steps

Research on DBS-fMRI has shown that the effects of DBS on blood oxygen level dependent (BOLD) signals are spread out, leading to the decision not to use global signal regression as it might lessen these signals. All preparation and analysis for the fMRI were conducted at the Welcome Centre for Human Neuroimaging at UCL, employing SPM12 software. The functional images underwent spatial smoothing with a Gaussian kernel sized at 6 mm full-width at half-maximum (FWHM) (see Fig. 2), had their slice timing adjusted, and were corrected for motion. They were carefully aligned with each participant's T1-weighted anatomical picture and were nonlinearly adjusted to fit the Montreal Neurological Institute (MNI) typical area [6].

The setup for collecting data for DBS-fMRI is illustrated in Fig. 1. (A) fMRI data was gathered from Parkinson's disease patients who had their brains stimulated on the left side, under both ideal and less optimal DBS circumstances. (B) During a data collection phase lasting 360 seconds, a 30-second DBS ON/OFF cycling setup was utilized.

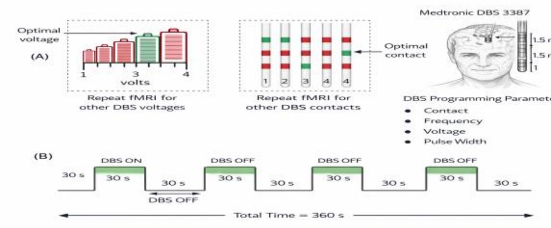

Fig. 1 DBS–fMRI Acquisition Paradigm

To reduce errors caused by patient movement, the Artifact Detection Tools (ART) toolbox was employed to spot and eliminate any volumes that showed motion exceeding 2 mm. Each patient could have up to 3. 3% of their total collected volumes discarded, which amounted to a maximum of six volumes. Additionally, six parameters that indicated rigid body movements (shifts in x, y, and z and rotations in yaw, pitch, and roll) were included as factors in the fMRI design model to address the motion effects. Correlation analysis showed that these motion variables were not significantly linked to the DBS ON/OFF cycling pattern, indicating that the observed BOLD variations resulted from stimulation rather than movements of the head. Fig 2 shows a data preparation workflow for fMRI using an autoencoder to extract features. The processed fMRI data produced statistical parametric t-maps, which were then inputted into the autoencoder to detect concealed features.

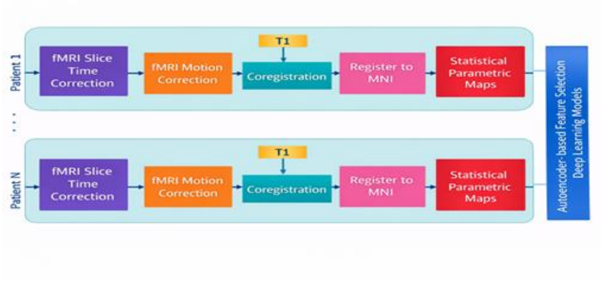

Fig. 2 fMRI Preprocessing and Autoencoder-Based Feature Extraction

A 30-second DBS ON/OFF block design was utilized to create statistical parametric maps (t-maps) through a general linear model. Following previous research that demonstrated its effectiveness in capturing DBS-induced BOLD responses across various brain areas and patients, the hemodynamic response function was modeled using a standard double-gamma function [7].

D. Process for Clinical Implementation

In actual medical scenarios, we offer a complete, step-by-step approach for semi-automated DBS optimization in one appointment to implement the recommended AE-MLP structure (Fig. 7). The four consecutive stages of the process are:

1) fMRI-based testing of DBS parameters Session: Patients engage in an fMRI session lasting 45 to 60 minutes using a 30-second ON/OFF block methodology. A range of systematically varied suboptimal configurations—such as too high or too low voltage, various frequencies, and different contact points—as well as the best medically determined choice are examined in a predetermined sequence. Every DBS equipment has to be MRI safe, and scanning must adhere to the safety guidelines set down by the producer.

2) Automated feature extraction and parameter estimation: The fMRI data acquired is analyzed in nearly real-time, as shown in Section III-C. The clever AE draws crucial features from every response map pertinent to DBS-fMRI without any human interaction. The MLP models next use these characteristics to estimate the ideal voltage, frequency, and contact points, as well as to categorize stimulation conditions as optimum or non-optimal.

3) Change and clinical evaluation: The movement disorder specialist can see the suggested parameters with an intuitive graphical interface. The clinician modifies recommendations as needed, evaluating them in light of particular patient characteristics, including anatomical abnormalities and side effects. This approach uses computational speed while ensuring clinical authority with human supervision included[8].

4) Testing and Enhancement: The same visit can verify the proposed ideal setting by means of basic clinical tests—finger tapping or gait analysis, for instance—to look for changes if it is controllable. Minor modifications can be made using immediate clinical input.

This process aims to cut the time spent programming for every patient from many months to under two hours without disrupting current clinical processes and without needing more training for experts.

E. Feature Learning with Autoencoders

To enhance the reliability of features obtained from DBS-fMRI response maps for classifying and predicting DBS parameters, we used a feature learning method based on an unsupervised autoencoder (AE). An AE is a type of neural network that performs unsupervised learning by identifying inherent relationships in the data. It does this by converting high-dimensional data into a simpler lower-dimensional form and then reconstructing the original data from that form. The AE has a balanced E encoder-decoder D setup, where the encoder transforms the input data x into a latent space z using adjustable weights, activation functions, and (w, δ, b) biases. The decoder then rebuilds the input using related parameters (w', δ', b').

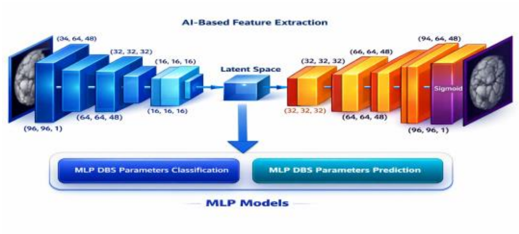

Fig. 3 AE–MLP Pipeline for DBS Parameter Optimization

The encoder in an autoencoder network transforms the input data into a hidden representation, which the decoder then uses to reconstruct the original input from this hidden space. The autoencoder's goal is to find mappings for the encoder and decoder that minimize the discrepancy between the actual data and what is recreated. This is accomplished by reducing a reconstruction loss, which reflects how well the output fits the original input. Based on the distribution of the input data, the average of this loss is determined. The autoencoder's parameters are optimized via backpropagation of gradients throughout the training phase. The mean squared error loss is commonly employed to assess the discrepancies between the input image and its reconstruction pixel by pixel in order to maintain simplicity and efficiency. Nevertheless, because it ignores perceptual features such as structural patterns and visual consistency, mean squared error fails to accurately capture how people see pictures. Researchers used a perceptual loss for the reconstruction loss that was based on the Structural Similarity Index Measure, or SSIM, in order to address this problem. Unlike simple pixel-only measures, which only look at brightness, contrast, and structural features, SSIM evaluates the similarity of pictures that are more in tune with how humans see the world. One less the SSIM score between the original picture and its reconstructed version is used to determine the SSIM-based loss. The autoencoder can generate images that are more meaningful and visually similar to the originals by optimizing this loss function [9].

A convolutional autoencoder network that operated in an unsupervised manner was employed to extract associated features from functional MRI response maps of deep brain stimulation without any prior knowledge. The encoder consists of six convolutional blocks, each of which includes a batch normalization, a rectified linear unit activation, and a convolution layer. These blocks slowly extract relevant features and spatial patterns from the incoming fMRI maps. The number of filters in the encoder blocks ranges from 1 in the first block to 256 in the final block. The initial convolutional block employs a little kernel with a stride of one, whereas the subsequent blocks use larger kernels with a stride of two for down sampling [10].

The decoder, which consists of five up sampling blocks, is responsible for recreating the input data. Every one of these blocks has batch normalization, a rectified linear unit activation, and a transposed convolution layer. The decoder blocks' filters decrease gradually, ending with 91 filters in the final layer. To ensure consistent up sampling, the same kernel size and stride are used across all transposed convolution layers. With this autoencoder design, the fMRI response maps are reduced from their original dimensionality to a small hidden representation of 256 by 3 by 3. All DBS-fMRI response maps were converted from regular MNI sizes to a standardized spatial resolution to ensure compatibility with the network. After being resized, the maps were normalized to a range between zero and one before the start of the training.

F. Classification and Prediction of DBS Parameters Using MLP

The multilayer perceptron (MLP) is a type of fully connected neural network known for its ability to approximate almost any smooth and measurable function effectively. More details about MLP structures can be found in earlier research. In this study, the MLP included eight hidden layers, with all layers being interconnected. Each of the hidden layers, aside from the eighth one, featured a linear transformation followed by a ReLU activation function. To help prevent overfitting, dropout layers with rates of 25%, 15%, 15%, and 15% were utilized after the first four sets of layers. The eighth layer functioned as a fully connected layer, linking 16 neurons to one output, which indicated the likelihood of optimal DBS parameters for classification and the predicted optimal DBS value for regression. A sigmoid activation function was used in the output layer for the classification purpose to generate normalized probabilities [11].

G. Model Development, Training, and Validation

Training Procedures for AE-MLP Model (Algorithm 1)

Input: (X, Y) DBS-fMRI training data

- Number of neurons in an autoencoder, g
- Number of hidden neurons in the MLP, p
- α being the rate at which one learns
- k is the total number of training rounds
- **Loss functions**: CSM (for Autoencoder), CMSE (for MLP)

Output:

- An improved AE-MLP model that attains the maximum classification accuracy

1) Set up the autoencoder (AE) with gg neurons in its bottleneck layer.

2) Train the autoencoder for kk epochs using the input data XX along with the CSM loss function.

3) Retrieve the latent features ZZ from the encoder that has been trained.

4) Set up the multilayer perceptron (MLP) classifier with pp hidden neurons.

5) For i=1i=1 to kk do

a) Train the MLP by using the latent representation ZZ and the labels YY, applying the CMSE loss function.

b) Adjust the network weights using a learning rate of αα.

c) Check the classification accuracy on the validation dataset.

d) Store the model weights if the current accuracy is better than the highest recorded accuracy.

e) End

6. Return the AE-MLP model that has the best validation accuracy.

We carried out the training and testing of the model utilizing Python 3. 8 and PyTorch 1. 13. 0. The training method for the AE-MLP, detailed in Algorithm 1, comprised two main phases. Initially, the autoencoder (AE) was trained for 250 cycles on the DBS-fMRI dataset using a loss function based on SSIM and the Adam optimizer with a learning rate of 1×10^{-4}. In the second phase, the AE, which had already been trained, was set to evaluation mode with its weights unchanged to gather hidden features from the nominal DBS-fMRI response maps. These features were then employed to develop the MLP models for classification and prediction to accurately estimate DBS parameters. The MLP models underwent training for 100 cycles using binary cross-entropy loss to handle classification. To tackle class imbalance, we adjusted the binary cross-entropy loss according to class frequencies. For comparison, we trained Support Vector Machine (SVM) and Convolutional Neural Network (CNN) models using the same features and evaluation method, utilizing mean squared error loss for prediction and the Adam optimizer with a learning rate of 1×10^{-3}. We assessed model performance through 5-fold stratified cross-validation, allocating 80% of the data for training and 20% for testing. The AE-MLP classification model determined whether DBS parameters were optimal or not, while the prediction model estimated the ideal contact location (x, y, z in MNI space), voltage, and frequency, implicitly learning the MNI coordinates of the Medtronic 3387 electrode contacts [12].

H. Hyperparameter Selection and Training

Parameter	AE	MLP
Learning rate	1×10−4	1×10^{-3}
Optimizer	Adam	Adam
Dropout rates	N/A	25%, 15%, 15%, 15%
Batch size	16	16
Epochs	250	100
Loss function	SSIM	BCE (classification), MSE(Regression)

Table. 2 Model Hyperparameters

The hyperparameters for the autoencoder (AE) and multilayer perceptron (MLP) models were chosen based on how stable they were in reaching convergence and how well they performed during validation on the training data. No adjustments were made for individual subjects. After making these selections, all hyperparameters remained unchanged and were used uniformly in every experiment to guarantee consistent results and an equal comparison between the models.

I. *Robustness Assessment of Feature Extraction Using Autoencoders*

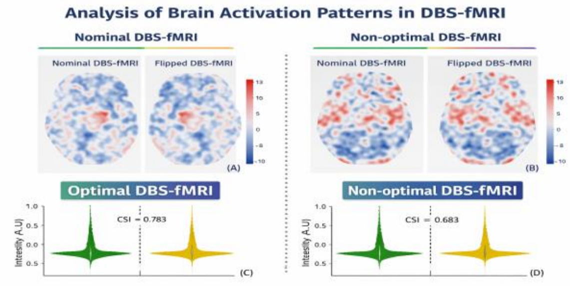

Fig. 4 Effect of Left–Right Flipping on DBS-fMRI Response Features

We assessed how well the autoencoder (AE) feature extraction model, which was only trained on left-sided DBS data, was able to learn distinct features from DBS-fMRI response maps that showed different activation patterns. To create variability based on stimulation direction and different disease states, we flipped the original left-sided (baseline) response maps from left to right, which rearranged the activated and deactivated areas and created 122 flipped maps. We then used the same pretrained AE to extract latent features from these flipped maps, allowing for a direct comparison with the features from the original maps. Furthermore, to evaluate the effect of including data from GPi patients, we trained separate classification and prediction models using only the nominal DBS-fMRI responses from STN patients (35 patients), keeping the same training and evaluation methods as those used for the combined STN-GPi models [13].

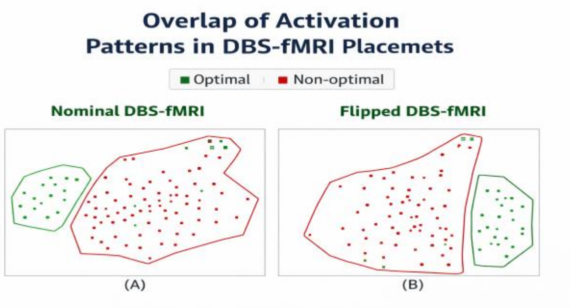

Fig. 5 t-SNE Clustering of AE-Extracted DBS-fMRI Latent Features

J. *Statistical Analysis & Visualization*

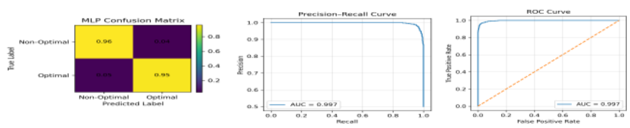

Fig. 6 Classification performance of the AE-MLP model: (A) confusion matrix, (B) precision–recall curve, and (C) ROC curve.

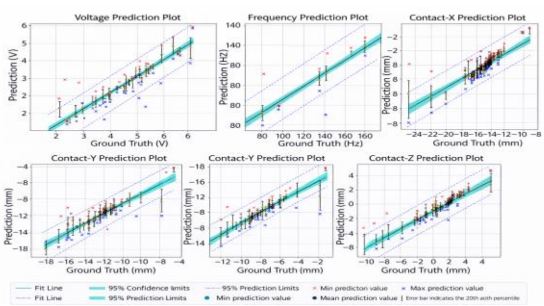

Fig. 7 Comparison between predicted and target optimal DBS stimulation parameters: (A) voltage, (B) frequency, and contact location in (C) x-, (D) y-, and (E) z- directions using the AE-MLP model.

DBS Parameter	Accuracy (10% Tolerance)	Accuracy (15% Tolerance)
Voltage (V)	0.79±0.04	0.91±0.03
Frequency (Hz)	0.85±0.04	0.89±0.04
Contract - X (mm)	0.82±0.05	0.94±0.04
Contract - Y (mm)	0.83±0.05	0.90±0.06
Contract - Z (mm)	0.70±0.07	0.76±0.08

Table. 3 AE-MLP Prediction Accuracy for DBS Stimulation Parameters

The cosine similarity index (CSI) and violin plots were used to analyze the latent characteristics extracted from the normal and changed DBS-fMRI response maps. The t-distributed stochastic neighbor embedding (t-SNE) approach was used to convert complex features into a two-dimensional visualization in order to show the latent features that the recommended AE feature extractor had identified from the 122 DBS-fMRI datasets. The accuracy, precision, recall, F1 score, receiver operating characteristic (ROC) curves, precision-recall curves, and area under the curve (AUC) were all used to assess the effectiveness of the AE-MLP classification model. The root mean square error (RMSE) and prediction accuracy within ±10% and ±15% of the actual values were used to evaluate the AE-MLP prediction model. The prediction results were also presented using linear regression plots, which showed mean expected values and their confidence intervals [14].

III. Experimental Results and Performance Analysis

Fig. 4 shows typical response maps for nominal and flipped DBS-fMRI along with the distribution of features derived from the AE for both optimal and non-optimal stimulation. Although the spatial activation patterns varied, the distributions of features were quite alike, with cosine similarity indices (CSI) recorded at 0. 783 for optimal and 0. 683 for non-optimal responses. The t-SNE analysis of AE-extracted features from the entire group of 122 samples revealed distinct clusters, successfully differentiating between optimal and non-optimal responses in both the nominal and flipped maps (Fig 5). The AE-MLP classification model, which utilized the combined STN–GPi dataset, reached an accuracy of 0. 96 ± 0. 04, a precision rate of 0. 92 ± 0. 07, a recall of 0. 95 ± 0. 07, and an F1 score of 0. 93 ± 0. 06.

This performance significantly exceeded that of the previously mentioned Parcel-Based Linear Discriminant Analysis (PB-LDA) method, which recorded an accuracy of 81%. The combined confusion matrix, ROC, and precision-recall curves are illustrated in Figure 6, showing AUCs of 0. 980 and 0. 976, respectively. The AE-MLP prediction model produced RMSE values of 0. 437 V, 12. 493 Hz, and 1. 405, 1. 241, and 1. 163 mm for voltage, frequency, and the x–y–z contact positions, as shown in Fig 7. The prediction accuracies at ±10% and ±15% tolerances are detailed in Table I, indicating higher accuracy at the higher 15% tolerance, with the z-coordinate prediction yielding the least accuracy [15].

Models developed using only the STN dataset showed similar performance in classification and predictions, which may be attributed to the small number of patients with GPi data. The STN-only classification model achieved 0. 97 ± 0. 02 accuracy, with AUCs of 0. 982 for ROC and 0. 980 for precision-recall. Meanwhile, the prediction model resulted in

RMSE values of 0. 517 V, 13. 93 Hz, and 1. 27, 1. 16, and 1. 21 mm for voltage, frequency, and x–y–z contact locations, as noted in Supplementary Figures S1–S2 and Table S1.

A. *Benchmarking Against Standard Models*

To better understand how our AE-MLP framework performs, we assessed it alongside two commonly used machine learning models that were trained on the same data set using the same 5-fold cross-validation method:

1) Support Vector Machine (SVM): We trained a linear SVM using flattened fMRI t-maps. In terms of classification, it reached an accuracy of 0. 87 ± 0. 05, precision of 0. 85 ± 0. 06, recall of 0. 82 ± 0. 07, and an F1-score of 0. 83 ± 0. 06. For regression, the RMSE values were 0. 62 V (voltage), 16. 8 Hz (frequency), and between 1. 78–2. 12 mm (for contact locations).

2) Convolutional Neural Network (CNN): A 5-layer CNN was trained from start to finish on fMRI maps for both classification and prediction tasks. The CNN achieved an accuracy of 0. 91 ± 0. 04, precision of 0. 89 ± 0. 05, recall of 0. 88 ± 0. 06, and an F1-score of 0. 88 ± 0. 05. Its regression RMSE values were 0. 51 V, 14. 2 Hz, and between 1. 45–1. 68 mm for contact locations.

Model	Accuracy	Voltage RMSE (V)	Freq RMSE (hz)	Contact RMSE (mm)
SVM	0.87 ± 0.05	0.62 ± 0.08	16.8 ± 2.1	1.78–2.12
CNN	0.91 ± 0.04	0.51 ± 0.06	14.2 ± 1.8	1.45–1.68
AE-MLP (Ours)	0.96 ± 0.04	0.44 ± 0.05	12.5 ± 1.5	1.16–1.41

Table. 3 Performance Comparison Across Models

Our AE-MLP framework surpassed both models in every metric (Table 4). Although the SVM was quick in computations, it struggled to capture complex spatial patterns in the fMRI data. The CNN had better results but was likely to overfit due to the small size of the dataset. The AE-MLP approach gained advantages from unsupervised pre-training, which helped refine feature learning and enhance generalization [16].

IV. RESULTS DISCUSSION

A. *Discussion of Methodology*

The growing emphasis on employing biomarkers to semi-automate DBS optimization is consistent with this study. This is accomplished by employing an unsupervised autoencoder to extract features from DBS-fMRI response maps and an MLP to categorize and forecast the optimum DBS parameters. Although it may be feasible to use the full DBS-fMRI response maps directly, their enormous size, complicates the calculation and complicates the interpretation of the findings. This emphasizes how crucial it is to have a solid method for feature extraction.

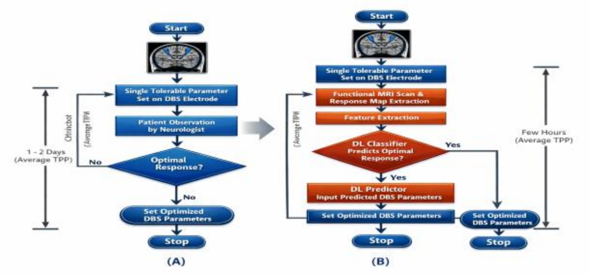

Fig. 8 Comparison of conventional empirical DBS programming and the proposed semi-automated fMRI-guided workflow with iterative classification and prediction of optimal stimulation parameters.

B. *Clinical Feasibility*

Fig 8 shows a suggested process for clinical application where DBS-fMRI is performed during one follow-up appointment after implantation. This is then complemented by automatic extraction of features, classification, and predicting parameters. The clinician is involved throughout, using the model's results to assist in targeted parameter testing instead of going through extensive empirical searches. This semi-automated method maintains clinical supervision while significantly cutting down programming time, making it easier to blend into current DBS clinical practices. The AE-MLP framework proved to be highly effective, with classification accuracy increasing from 81% with the earlier PB-LDA approach to 96%, underlining the importance of good feature learning. The prediction model's RMSE values were within clinically accepted ranges; particularly, the error in voltage prediction (0. 437 V) aligns with standard clinical adjustment increments. These outcomes endorse the practicality of this method for quick semi-automated DBS programming [17].

One AE model trained with the left-sided DBS-fMRI data successfully separated the best and least effective stimulation features in both normal and left-right flipped response maps, as shown by t-SNE visuals. Even with different activation patterns, the features extracted by the AE remained stable, indicating they are reliable despite changes in space. While flipping left and right does not completely mirror actual variations due to the stimulation side or condition, these results imply that AE-driven feature extraction is strong against these changes, and flipping could be an effective method for data enhancement. The DBS-fMRI approach used brief stimulation cycles to observe the blood flow responses linked to DBS, while long-term clinical results served as the benchmark. Even though immediate and long-term responses to DBS can vary, previous studies have shown that this fMRI method consistently differentiates optimal from suboptimal DBS settings across various conditions, reinforcing its potential as a swift biomarker for optimizing DBS.

C. *Restrictions*

It's important to take into account a number of significant constraints. Although the dataset had an unbalanced distribution, with more non-ideal responses than ideal ones, the models were able to handle this imbalance well, as evidenced by the high F1 scores and the tight clustering of features from the AE extraction. In addition, the limited number of individuals with GPi may limit the ability to generalize conclusions for that population. Due to limited access to DBS-fMRI data, the total size of the dataset was constrained, which may impact the performance of more recent DBS electrodes that have more parameters. It is hoped that ongoing initiatives to collect more data, particularly from directional electrodes and various pulse-width configurations, will improve the models' accuracy and reliability[18].

Despite these restrictions, the proposed AE-MLP technique demonstrates potential for reducing DBS programming time from months or years to just a few hours during a single physician appointment. For disorders with sluggish or difficult-to-assess responses, such a semi-automated fMRI-based DBS programming framework would be revolutionary. Additionally, it would reduce the workload in clinical environments while promoting the wider use of sophisticated DBS technologies. The lack of an external validation group also restricts the capacity for generalization. Future research will focus on acquiring data from several centers and prospectively validating it to ensure that the findings are consistent across various machines, surgical facilities, and patient populations, even though a stratified cross-validation approach was used to avoid overfitting [19].

D. Safety and Feasibility Considerations

When moving forward with the clinical use of fMRI-guided DBS optimization, it's crucial to prioritize safety, compatibility, and challenges tied to practical execution:

1) MRI Safety and Artifact Management: All DBS systems that are implanted need to be safe for MRI use, and the scanning process must follow the specific rules set by the manufacturer regarding magnetic field strength, specific absorption rate, and gradient switching limits. In this research, we utilized Medtronic Activa PC/RC devices that have been labeled as MRI-safe for both 1. 5T and 3T scans. To lessen imaging errors, we used spin-echo EPI sequences, which help to minimize problems near electrode contacts. We also employed post-processing techniques, like the ART toolbox, to detect and eliminate any areas with significant signal loss or distortion[20].

2) Real-World Clinical Integration: The suggested system is meant to work well with typical clinical MRI machines and DBS programming setups. Still, there are a few practical issues to consider:

a) adding fMRI scans can make the clinical visit 45 to 60 minutes longer, potentially affecting patient comfort and clinic efficiency.

b) patients suffering from strong tremors or anxiety may struggle with prolonged scanning.

c) scheduling needs to account for the necessary medication adjustment before scanning. Future steps might look into shorter fMRI procedures or resting-state scans to cut down scan duration.

3) Regulatory and Validation Process: Before this system can be used clinically, it needs regulatory approval (like from the FDA or for CE marking) as software that serves as a medical device. This involves multi-center trials that confirm both safety and effectiveness in comparison to standard practices. Moreover, the system's performance must be verified across different types of equipment (such as various DBS electrodes and MRI machines) and patient groups (with different levels of illness and other health issues).

4) Ethical and Understanding Factors: Though the AE-MLP pipeline can provide numerical recommendations, the final choice about programming should always rest with the clinician. Therefore, the system must integrate features that help explain its decisions—such as visual maps showing brain regions that affect its predictions—to ensure clarity and foster trust[21].

Even with these hurdles, the chance to cut down DBS optimization time from months to just a few hours make it worthwhile to keep developing and validating this method. Upcoming efforts will be aimed at simplifying the process, enhancing resistance to errors, and running clinical trials in the future [22].

V. Conclusion

Using an AE approach for automated feature extraction from DBS-fMRI response maps taken during left-sided DBS in Parkinson's disease patients, we established a framework based on prior research indicating that fMRI is a trustworthy predictor for the ideal DBS settings. Although there were changes in the disease condition or the side of stimulation, the characteristics acquired by AE were able to successfully distinguish between successful and unsuccessful stimulation, demonstrating their reliability. These features improved the classification and prediction of DBS parameters using an MLP approach, which was much superior to the PB-LDA approach that had been used before. Our suggested AE-MLP framework offers a lot of potential for partially automating the optimization of fMRI-guided DBS, cutting the amount of time needed for parameter adjustment from around a year to only a few hours during a single clinic appointment.

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